CROP YIELD & WEATHER ANALYSIS — INDIA

An interactive Python dashboard for exploring crop and climate patterns

PROJECT PRESENTATION • OCTOBER 2026

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Project at a Glance

A quick overview of the project and its purpose

The Project

Explore historical crop yield, production and weather patterns across Indian districts.

The Data

District-wise crop and climate records covering 1984–2017.

The Output

An interactive dashboard with filters, charts, summary statistics and CSV export.

PROJECT WORKFLOW

01 **Upload CSV** — 02 **Select & filter** — 03 **Explore charts** — 04 **Download data**

Problem Statement & Objectives

Making large agricultural datasets easier to explore

Problem

• Agricultural records span many years, states, districts and crops.

• Manual analysis can be slow and makes comparisons harder.

• Users need a simple way to explore how crop measures vary with rainfall and temperature.

Objectives

• Visualize year-wise crop yield or production trends.

• Compare selected crops and districts.

• Explore rainfall–yield and temperature–yield relationships.

• Offer interactive filters, summary statistics and CSV downloads.

Key question: What patterns can be observed in crop productivity and weather across years and regions?

Dataset Overview

All-India District-wise Crop and Climate Dataset (1984–2017)

242,696
records

32
columns

26
numerical columns

1984–2017
time period

Agricultural variables

Year • State • District • Crop • Area • Production • Yield • Season

Weather variables

Rainfall / precipitation • Maximum and minimum temperature • Wind speed • Relative humidity • Solar radiation

Format: CSV | Data source listed in the report: Kaggle dataset

Methodology & Data Preparation

From raw records to meaningful visual exploration

1. **Load** — Upload CSV and read it with Pandas.
2. **Check** — Inspect columns, data types, missing values and duplicates.
3. **Prepare** — Handle data issues carefully; engineer features when needed.
4. **Analyze** — Use summaries, comparisons, correlation and statistical tests.
5. **Present** — Display results through charts and an interactive Streamlit dashboard.

Tools: Python • Pandas • NumPy • Matplotlib • Streamlit

Exploratory Data Analysis

Use visualizations and statistics to inspect patterns

Univariate analysis

Histograms and boxplots show distributions, variability and potential outliers.

Bivariate analysis

Scatter plots compare a weather variable with crop yield or production.

Correlation analysis

A heatmap summarizes the strength and direction of linear relationships.

Variables explored

- Crop yield
- Production
- Rainfall
- Temperature
- Area

Important: correlation can show association, but it does not prove that weather caused a change in yield.

Interactive Dashboard Features

Built with Streamlit for accessible data exploration

CSV upload

Load the district-wise dataset.

Column selection

Map state, district, crop, year, yield and weather fields.

Filters

Focus on a state, district or crop.

Summary statistics

Review record counts and average selected measures.

Charts

View year trends, district comparisons and weather scatter plots.

CSV export

Download the filtered records for further work.

The dashboard supports exploration; it does not currently predict future crop yields.

Insights & Recommendations

What the dashboard helps users investigate

Analysis opportunities

- Identify changes in crop measures across 1984–2017.
- Compare crop productivity across districts and crop types.
- Explore possible rainfall and temperature associations.
- Examine regional differences using filters.

Recommendations

- Use reliable and updated agricultural data.
- Consider weather patterns alongside crop records.
- Investigate districts with consistently low recorded yield.
- Validate charts and statistical outputs before making decisions.

Specific numerical findings should be reported only after checking the actual dashboard outputs.

Limitations & Future Scope

Current boundaries and possible improvements

Current limitations

- Historical dataset ends in 2017.
- Analysis depends on data quality and available weather columns.
- Correlation does not establish cause and effect.
- The dashboard does not guarantee future-yield predictions.

Future improvements

- Integrate recent agricultural data and weather forecasts.
- Explore machine-learning yield prediction models.
- Add interactive maps for district-level patterns.
- Include soil, irrigation and fertilizer variables where reliable data exist.

Goal: make agricultural data easier to understand and support further research.

THANK YOU!

Crop Yield & Weather Analysis — India

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